FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2021

TEST DATE: 14-02-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $\vec{V} = 2t\,\hat{i} + t^2\,\hat{j}$

$\vec{a} = 2\hat{i} + 2t\hat{j}$

$\left|\vec{V}\right| = V = \sqrt{4t^2 + t^4}$

$a_t = \left(\frac{dV}{dt}\right) = \frac{1}{2\sqrt{4t^2 + t^4}} \times \left(8t + 4t^3\right)$

$at\ t = 1\ a_t = \left(\frac{6}{\sqrt{5}}\right) m/s^2$

$a = \sqrt{2^2 + (2t)^2} = \sqrt{8} m/s^2$

$r = \sqrt{5}$

$a_c = \sqrt{a^2 - a_t^2} = \sqrt{8 - \frac{36}{5}} = \left(\frac{2}{\sqrt{5}}\right)$

$R = \left(\frac{V^2}{a_c}\right) = \frac{5}{\left(\frac{2}{\sqrt{5}}\right)} = \frac{5\sqrt{5}}{2}$

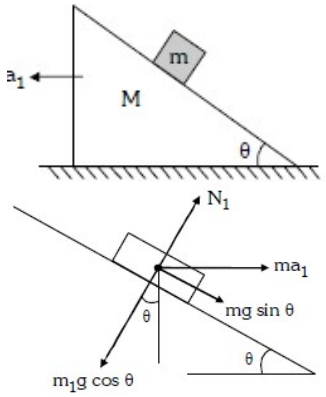
$R = \left(\frac{5\sqrt{5}}{2}\right)\text{m}$

2. D

Sol. For entire system, $\vec{F}_y = (M+m)a_{cmy}$

$(M+m)g - N = (M+m)a_{cmy}$

$N = (m+M)(g - a_{cmy})$ _____(1)



$N_1 + ma_1 \sin\theta = mg\cos\theta$

$N_1 = m[g\cos\theta - a_1 \sin\theta]$

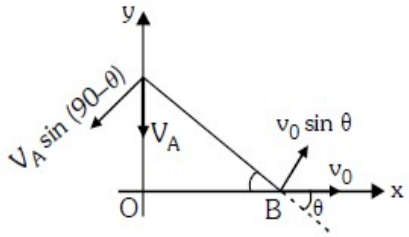
3. B

Sol. $V_A \cos(90-\theta) = V_0 \cos\theta$

$V_A \sin\theta = V_0 \cos\theta$

$V_A = \dfrac{V_0}{\tan\theta}$

$V_A = \dfrac{4}{3}V_0$ $\Rightarrow \omega = \dfrac{[V_A \sin(90-\theta) + V_0 \sin\theta]}{\ell}$



4. B

Sol. $F(2S) = \dfrac{1}{2}mv^2 + \dfrac{1}{2}I_{cm}w^2$

5. A

Sol. $\vec{L} = \vec{r} \times \vec{P}_{cm} + I_{cm}\vec{\omega}$

6. C

Sol. $VT = C_0$

$PV^2 = C_0,\ x = 2$

$C = C_v + \dfrac{R}{1-r} = \left(\dfrac{3R}{2}\right)$

$W = \left[\dfrac{nR\Delta T}{(-x)}\right] = \dfrac{2R \times 200}{(1-2)} = -400R$

7. B

Sol.

	Q	ΔU	W
AB	–70	0	–70
BC	150	150	0
CA	60	-150	210

$\eta = \dfrac{140}{210} \times 100\% = 60\%$

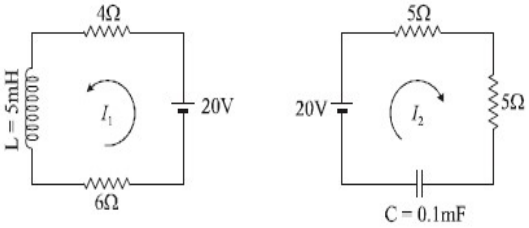
8. B

Sol. $I_1 = \frac{20}{10}\left(1 - e^{\frac{-t}{5\times10^{-4}}}\right) = 2\left(1-\frac{1}{4}\right) = \frac{3}{2} = 1.54$

$I_2 = \frac{20}{10}e^{\frac{-t}{10^{-3}}} = 1A$

From superposition

$I = I_1 + I_2 = 2.5A$



9. A

Sol. The acceleration of point of contact is $\left(\frac{V^2}{R}\right)$ towards centre. The a_{cm} and friction may or may not be zero.

10. C

11. B

12. C

13. C

14. B

15. A

16. C

17. D

18. B

19. B

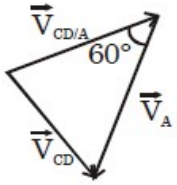
20. D

Sol. $\vec{V}_{CD/A} = \vec{V}_{CD} - \vec{V}_A$

$\vec{V}_{CD} = \vec{V}_{CD/A} + \vec{V}_A$

$\vec{V}_{CD}^2 = \vec{V}_A^2 + \vec{V}_{CD/A}^2 - 2V_A V_{CD/A}\cos 60°$

$V_{CD} = 2\,m/s$



SECTION – B

21. 00008.00

Sol. Process $A \to B$

$W_{AB} = \int Pdv = \int \frac{3}{2}T^{1/2}dv$

$= \int \frac{3}{2}T^{1/2} \times \frac{1}{3}RT^{-1/2}dT$

On solving, $W_{AB} = 50\ R = 50 \times 8.3 = 415J$

Process $B \to C$

$U = \frac{1}{2}V^{1/2}$

$\frac{3}{2}RT = \frac{1}{2}V^{1/2} \Rightarrow 3PV^{1/2} = 1$

$\therefore P = \frac{1}{3\sqrt{V}}$

Now $W_{BC} = \int Pdv = \int_{100}^{1600} \frac{1}{3\sqrt{V}}dv = \frac{3}{2}\sqrt{V}$

$= \frac{2}{3}\left[40 - 10\right] = \frac{2}{3} \times 30 = 20J$

Total $W = 415 + 20 = 435$

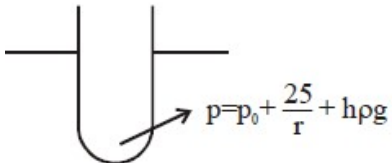
22. 00008.00

23. 00006.00

Sol. $\Delta p = \frac{25}{r} + h\rho g$

$= \frac{2 \times 7 \times 10^{-3}}{0.35 \times 10^{-3}} + 2 \times 10^{-2} \times 10^{4} = 240$

$\frac{\Delta p}{40} = 6$



24. 00002.00

Sol. $T - mg = ma_b = ma_m$

$2T - 2mg = 2ma_B$

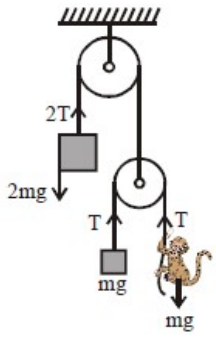
$\Rightarrow a_B = a_b = a_m \Rightarrow v_B = v_b = v_m = v$

$v_p = -v$

$v_R - v_b = 2v_p$

$v_R = -2v - v = -3v$

$v_{mR} = v - (-3v) = 4v = 8$



25. 00002.00

26. 00005.00

Sol. Let X' be the leftward displacement of A and x and y be the leftward and downward displacements of m. Then by constraint

$x = X \Rightarrow x = X \Rightarrow a_x = A_x$

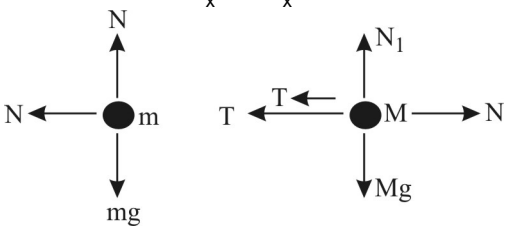
and $l_1 - x + l_2 + l_3 - x + l_4 + y = l_1 + l_2 + l_3 + l_4$

where l_1, l_2, l_3, l_4 are the instantaneous lengths of the segments of the string

$\Rightarrow 2x = y \Rightarrow 2x = y \Rightarrow 2a_x = a_y$

$N = ma_x$ and $mg - T = ma_y$

and $2T - N = MA_x = Ma_x$



Eliminating T, A_x and N

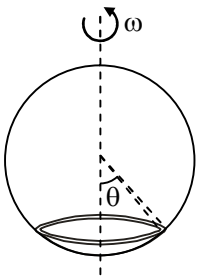
$a_x = \frac{2mg}{M+5m}$ and $a_y = \frac{4mg}{M+5m}$

$a = \sqrt{a_x^2 + a_y^2} = \frac{2\sqrt{5}mg}{M+5m}$

27. 00003.00

Sol. When the shell rotates, current is induced due to motion of charge. To calculate magnetic induction at centre of the shell, rotating shell can be assumed to be composed of thin circular current carrying rings. Such a ring can be assumed as follows:

Consider a radius of the shell inclined at angle 'q' with the axis of rotation. This radius is rotated about the axis keeping q constant. Thus a circle is traced as shown in fig.



Its radius, $r = R \sin q$

Distance of its centre from centre of the shell, $x = R \cos q$.

Now consider another radius inclined at angle (q + dq). It is also rotated in the same way and another circle is traced. The portion between two circles forms a circular ring.

Area of this ring = $2\pi r R\, dq = 2\pi R^2 \sin q\, dq$

Charge on this ring, $dq = s \cdot 2\pi R^2 \sin q\, dq$

Since, angular velocity of the shell is w, therefore, it completes $\frac{\omega}{2\pi}$ revolutions per second.

Hence, current associated with the ring considered,

$i = \frac{\omega}{2\pi} dq = s\, wR^2 \sin q\, dq$

Since, centre of the shell is a point lying on the axis of a circular coil of radius r, carrying current i at a distance x from centre of the coil, therefore, magnetic induction at centre of the shell due to this coil is

$dB = \frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}}$

$= \frac{1}{2} \mu_0\, s\, wR \sin^3 q\, dq$

Hence, resultant magnetic induction at centre of the shell

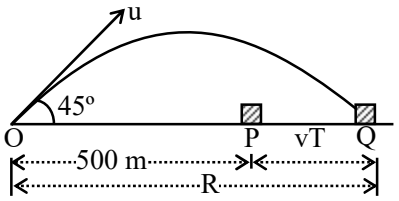
$B = \int dB = \frac{1}{2} \mu_0\, s\, wR \int_0^{\pi} \sin^3 \theta\, d\theta = \frac{2}{3} \mu_0\, s\, wR$

28. 00088.55

Sol. Consider that at the instant, the shell is fired from the gun at point O, the car is at point P The shell will move along parabolic path for T its time of flight and will then strike the ground at point Q at a distance R, equal to the horizontal range. Therefore, the shell will hit the car, if in the time T (time of flight), the car moves the distance PQ= v T, where v is velocity of the car. Since the car is initially at a distance of 500 m from the gun, it follows that

$R = 500 + v\,T$ .....(1)

(a) Let u be the speed of projection of shell.



Now, $R = \frac{u^2 \sin 2\theta}{g}$

Here q = 45°

$R = \frac{u^2 \sin 2\times 45^\circ}{9.8} = \frac{u^2 \sin 90^\circ}{9.8} = \frac{u^2}{9.8}$

Also, $T = \frac{2u\sin\theta}{g} = \frac{2u\sin 45^\circ}{9.8}$

$= \frac{2u}{9.8} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{2}u}{9.8}$

Further, velocity of the car, v = 72 km h^{-1}

= 72 × (1000m) × (60 × 60 s)$^{-1}$ = 20 m s^{-1}

Substituting for R, T and v in equation (1), we have $\frac{u^2}{9.8} = 500 + 20 \times \frac{\sqrt{2}u}{9.8}$

or $u^2 - \sqrt{2}\,u - 500 \times 9.8 = 0$

On solving for u we have

$u = 10\,(\sqrt{2} + \sqrt{51})$

u = 10 (1.414 + 7.141)

= 10× 8.555 = 85.55 m s^{-1}

29. 00120.00

Sol. As here initially

$P_I = P_0 = 1 \times 10^5$ N/m^2,

$V_I = V_0 = 2.4 \times 10^{-3} m^3$,

$T_I = T_0 = 300K$

and finally

$P_F = P_0 + \frac{Kx}{A} = 1 \times 10^5 + \frac{8000 \times 0.1}{8 \times 10^{-3}}$

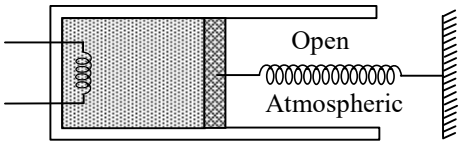
$= 2 \times 10^5$ N/m^2,

$V_F = V_0 + Ax = 2.4 \times 10 + 0.1 \times 8 \times 10^{-3}$

$= 3.2 \times 10^{-3} m^3$,

So from gas equation PV = μRT, i.e.

$\frac{P_I V_I}{T_I} = \frac{P_F V_F}{T_F}$



we have

$T_F = \frac{P_F}{P_I} \times \frac{V_F}{V_I} \times T_I$

$= \frac{2 \times 10^5}{1 \times 10^5} \times \frac{3.2 \times 10^{-3}}{2.4 \times 10^{-3}} \times 300 = 800$ K

As DW = ∫ PdV,

here P = P_0 + (kx/A)

so DW = $\int_0^{0.1} \left(P_0 + \frac{kx}{A} \right) A \, dx$

$= \int_0^{0.1} (P_0 A + kx) dx$

i.e. = $P_0 Ax + ½ kx^2$

= [$10^5 \times 8 \times 10^{-3} \times 0.1 + ½ \times 8000 \times (0.1)^2$]

i.e. DW = 80 + 40 = 120J

30. 00680.00

Sol. The observer A is at rest with respect to the air and the source is travelling at a velocity of 120 km/h i.e., 100/3 m/s. As is clear from the figure, the person receives the sound of the whistle in a direction BA making an angle q with the track where cos q = 300/500 = 3/5. The component of the velocity of the source (i.e., of the train.) along with this direction AB is $\frac{100}{3} \times \frac{3}{5}$ m/s = 20m/s.

As the source is approaching the person with this component, the frequency heard by the observer is.

$n' = \frac{v}{v - v\cos\theta} \nu = \frac{340}{340 - 20} \times 640\text{Hz} = 680$ Hz.

Chemistry PART – II

SECTION – A

31. B

32. A

33. C

34. D

35. C

36. B

37. B

38. B

39. D

40. C

41. B

42. A

43. B

44. C

45. C

46. C

47. C

48. B

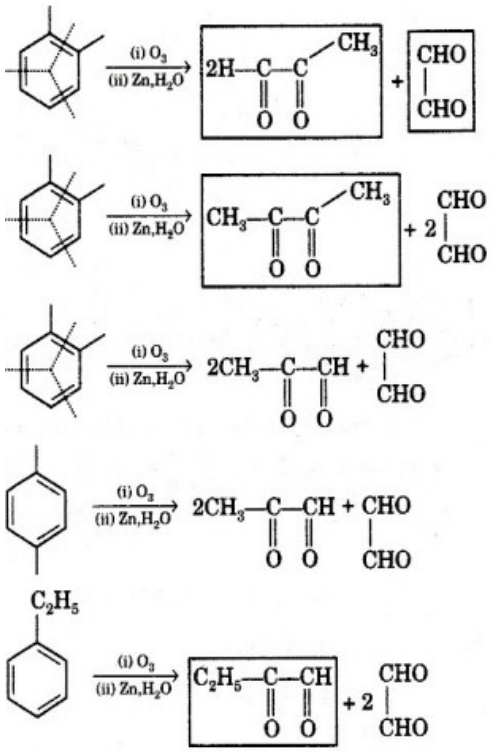
49. C

50. A

SECTION – B

51. 00004.00

Sol.



52. 00007.00

Sol. except (iii) all other reactions can form esters

53. 00001.17

Sol. $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

Since NH_4HS is a solid, hence $a_{NH_4HS} = 1$ and its un-dissociated amount will not effect the total pressure (due to gaseous NH_3 and H_2S only). Let x be its moles decomposed at equilibrium and P be the equilibrium pressure.

Moles	NH_4HS	NH_3	H_2S
Initial	a	0	0
At equilibrium	$a-x$	x	x

Total moles at equilibrium = Moles of $(NH_3 + H_2S) = 2x$ (only gaseous moles)

$P = ?$ $\quad K_p = 0.34$

$$P_{H_2S} = \frac{x}{2x}P = \frac{P}{2} \text{ and } P_{NH_3} = \frac{x}{2x}P = \frac{p}{2}$$

(For equimolar proportions, partial pressure are equal)

$$\therefore K_p = P_{H_2S}.P_{NH_3} \quad (a_{NH_4HS} = 1)$$

$$\Rightarrow 0.34 = \frac{P}{2} \times \frac{P}{2}$$

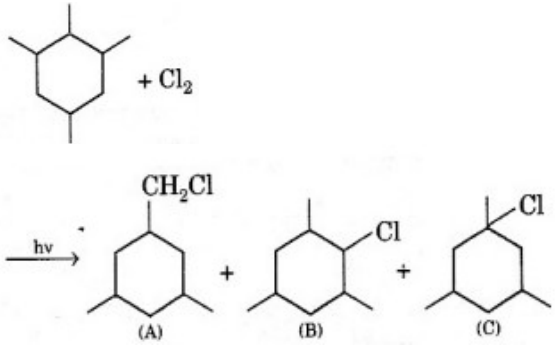
$$\Rightarrow \frac{P^2}{4} = 0.34 \Rightarrow P = \sqrt{4 \times 0.34} = 1.17 \text{ atm}$$

$\Rightarrow P_{NH_3} = \frac{P}{2} = \frac{1.17}{2} = 0.585\,atm$

And $P_{H_2S} = \frac{P}{2} = \frac{1.17}{2} = 0.585\,atm$

54. 00050.33

Sol.



Relative reactivity of $1°:2°:3°\,H$ towards photochlorination as calculate by above % yield is $1:3.8:4.5$. So, yield of A, B and C will be ratio

$9\times1\ :\ 6\times3.8\ :\ 3\times4$ i.e., $9:22.8:13.5$

So, major product is B and its % yield is $\frac{22.8}{45.3}\times100 = 50.33$

55. 02870.80

Sol. $\Delta_f H_{CaCl_2} = \Delta_{sub}H_{Ca} + \Delta H_{Cl-Cl} + \Delta_f H_{Ca^{2+}} + 2\times\Delta_{eg}H_{Cl} + \Delta_l H_{CaCl_2}$

Or $-795 = 121 + 242.8 + 2422 - (355\times2) + \Delta_l H_{CaCl_2}$

$\Delta_l H_{CaCl_2} = -2870.8\,kJ\,mol^{-1}$

56. 00061.30

Sol. At constant volume and at 347°C moles $\propto$ pressure $a \propto 31$ before equilibrium

$a + 2x \propto 50$ after equilibrium

$\therefore \frac{a+2x}{a} = \frac{50}{31} \therefore x = \frac{19}{62}a$

$\therefore$ % of NH_3 decomposed $= \frac{2a}{a}\times100 = \frac{2\times19a}{62\times a}\times100 = 61.3\%$

57. 00041.36

Sol. $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g);\ \Delta_r G = -33.32\,kJ$

Using: $\Delta_r G = \Delta_r G^{\ominus} - RT\ln Q$

Where $Q = \frac{p^2_{NH_3}}{p_{N_2}p^3_{H_2}} = \frac{1^2}{1\times3^2} = \frac{1}{27}$

$\Rightarrow \Delta_r G = -33.2 + \left(8.314\times10^{-3}\right)\times298\times2.303\log_{10}\frac{1}{27}$

$\Rightarrow -33.2 - 8.16 = -41.36\,kJ\,mol^{-1}$

58. 00010.00

Sol. $CaOCl_2 \equiv Cl_2 \equiv I \equiv S_2O_3^{2-}$

$mEq \equiv mEq \equiv mEq \equiv mEq$

Weight of Cl_2 = mEq × $10^{-3} \times Ew(Cl_2)$

$$= 2 \times 10^{-3} \times \frac{71}{2}$$

$$= 0.071\,g$$

% of available $Cl_2 = \dfrac{Weight\ of\ Cl_2}{Weight\ of\ CaOCl_2} \times 100$

$$= \frac{0.071 \times 100}{0.71} = 10\%$$

59. 00189.79

Sol. B. pt. of benzene $= 80 + 273 = 353K$

$$\therefore\ K_b = \frac{RT^2}{1000l} = \frac{2 \times (353)^2}{1000 \times 90} = 2.77\ K\ mol^{-1}\ kg$$

$\Delta T = 80.175 - 80 = 0.175, w = 83.4\,g$

$$0.175 = \frac{1000 \times 2.77 \times 1}{m \times 83.4}$$

$m = 189.79$

60. 00001.84

Sol. Given: $Fe^{6+} + 3e \longrightarrow Fe^{3+}$; $\quad -\Delta G_1^\circ = 3 \times 2.20 \times F \quad .....(i)$

$Fe^{3+} + e \longrightarrow Fe^{2+}$; $\quad -\Delta G_2^\circ = 1 \times 0.77 \times F \quad .....(ii)$

$Fe^{2+} + 2e \longrightarrow Fe^{0}$; $\quad -\Delta G_3^\circ = 2 \times (-0.445) \times F \quad .....(iii)$

By Eq. (i) and (ii)

$Fe^{6+} + 4e \longrightarrow Fe^{2+}$; $\quad -\Delta G_4^\circ = -\Delta G_1^\circ - \Delta G_2^\circ$

$\because \quad 4 \times E_4^\circ \times F = 3 \times 2.20 \times F + (1 \times 0.77 \times F)$

$$E_4^\circ = 1.84\ \text{V}$$

Mathematics

PART – III

SECTION – A

61. A

Sol. Let $A=\int_{a-1}^{a+1} e^{-(x-1)^2}dx$

$\frac{dA}{da}=e^{-a^2}-e^{-(a-2)^2}$

For maximum or minimum $\frac{dA}{da}=0$

We get $a=1$ & $\left(\frac{d^2A}{da^2}\right)_{a=1}=-ve$

62. C

Sol. $\frac{e^{\ln x}.\operatorname{sgn}\left(x^2+x+1\right)}{\ln x}=k \qquad \Rightarrow \frac{x}{\ln x}=k$

$\therefore k\in(-\infty,0)\cup\{e\}$

63. D

Sol. $\sum_{n=1}^{\infty}\frac{3n^2+3n+1}{n^3(n+1)^3}$

$=\sum_{n=1}^{\infty}\left(\frac{1}{n^3}-\frac{1}{(n+1)^3}\right)=1$

64. D

Sol. As, $g(x)=f(x-\pi)+f(x+2\pi)$

$=\sin(x-\pi)+|\sin(x-\pi)|+\sin(x+2\pi)+|\sin(x+2\pi)|$

$=2|\sin x|$

So, $\int_{-2\pi}^{5\pi} g(x)dx=7\int_{0}^{\pi} g(x)dx=28$

65. B

Sol. $f'(x)=3x^2+2ax+a$

We have $\alpha+\beta=\frac{-2a}{3};\ \alpha\beta=\frac{a}{3}$ & $f(\alpha)+f(\beta)=2$

$\Rightarrow \alpha^3+\beta^3+a(\alpha^2+\beta^2)+a(\alpha+\beta)=$

On solving $a=\frac{9}{2}$

66. D

Sol. $I=\frac{1}{2}\int_0^{\pi/2} x\left|\cos 2x\right|dx$ put 2x=t

$I=\frac{1}{8}\int_0^{\pi/2} t\left|\cos t\right|dt \quad .....(1)$

$I=\frac{1}{8}\int_0^{\pi/2} (\pi-t)\left|\cos t\right|dt \quad .....(2)$

$2\mathrm{I}=\frac{\pi}{8}\int_0^{\pi}\left|\cos t\right|dt=\frac{\pi}{8}\times 2$

$\Rightarrow \mathrm{I}=\frac{\pi}{8}$

67. B

Sol. $x^2=e^{|x|-2}$

Draw graph.

68. B

Sol. $O=3, M=1, C=1, Y=1, N=2, T=2, I=2$

$\frac{9!}{2!\cdot 2!\cdot 2!}\times {}^{10}C_3=15\times 9!$

$\Rightarrow k=15$

69. C

Sol. $0,\ \pi,\ 2\pi,\ \frac{\pi}{3},\frac{2\pi}{3},\frac{4\pi}{3},\frac{5\pi}{3}$ as given expression $=\tan 3x\tan 2x\tan x$

Note: General solution is $\frac{4\pi}{3}$

70. A

Sol. $T_{f+1}=15C_r x^f\left(1+x\right)^f$

$r=3,\ 4,\ 5$

71. C

Sol. Let $\frac{\pi}{7}=\theta$

$\therefore\ 4\cos\theta\cos 2\theta-2\cos 2\theta$

$=\frac{4\cos\theta\cos 2\theta}{\sin\theta}\sin\theta-2\cos 2\theta$

$=\frac{\sin 4\theta-2\sin\theta\cos 2\theta}{\sin\theta}$

$=\frac{\sin 4\theta-\left(\sin 3\theta\sin\theta\right)}{\sin\theta}$

$\pi=7\theta$

$\therefore \sin 4\theta = \sin 3\theta$

$= \dfrac{\sin 3\theta - \sin 3\theta + \sin\theta}{\sin\theta} = 1$

72. D

Sol. Let 1st term be 'a' and common difference be 'd'

$\dfrac{10}{2}\{2a+9d\} = 80$

$2a + 9d = 16$

$a + 2d = 7$

$d = \dfrac{2}{5}$

$a = \dfrac{31}{5}$

$a_9 = a + 8d = \dfrac{47}{5}$

$[a_9] = 9$

73. B

Sol. $(A-B)^3 = A^3 - B^3 - 3A^2B + 3AB^2$

$= A - B - 3AB + 3AB$

$(A-B)^3 = (A-B)$

$|A-B|^3 = |A-B|$

$\Rightarrow |A-B| = 0$

74. A

Sol. Consider $z^2 = -|z|$

$|z| = 0 \ or \ |z| = 1$

$\Rightarrow z = 0, z^2 = -1, z = \pm i$

Required equation $x(x^2+1) = 0$

$x^3 + x = 0$

$a = 1, \ b = 0$

75. B

Sol. n(S) = number of ways to obtain total 2, 4, 6 or 8

$\Rightarrow$ n(S) = 4 + 9 + 2 $\times$ 3 + 1

n(E) = 9

p(E) = $\dfrac{9}{20}$.

76. B

Sol. Let tangent for parabola $y^2 = 4ax$ is $y = mx + \dfrac{a}{m}$

$\Rightarrow m^2x - my + a = 0 \qquad \Rightarrow m_1 + m_2 = \frac{y}{x}$, $m_1m_2 = \frac{a}{x}$

Then $\tan 45^\circ = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right| \quad \Rightarrow \left(1 + \left(\frac{a}{x}\right)\right)^2 = \left(\left(\frac{y}{x}\right)^2 - \frac{4a}{x}\right)$

Locus is $x^2 - y^2 + 6ax + a^2 = 0$

$\Rightarrow (x + 3a)^2 - y^2 = 8a^2$

$\Rightarrow$ Rectangular Hyperbola.

$\Rightarrow$ Centre is $(-3a, 0)$.

77. A

Sol. $x^2 - 1 \geq 0 \Rightarrow x \in (-\infty, -1] \cup [1, \infty)$

and $x^2 + x + 1 \in [1, \infty)$ for $x \in (-\infty, -1] \cup [1, \infty)$

given expression is defined only for $x = -1$.

Then $\sin^{-1}1 + \cos^{-1}0 = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

$\Rightarrow$ No solution.

78. B

Sol. Let $(a\cos\theta, a\sin\theta)$ be a point on $x^2 + y^2 = a^2$, whose chord of contact with respect to the circle $x^2 + y^2 = b^2$ is $ax\cos\theta + ay\sin\theta = b^2$, which touches $x^2 + y^2 = c^2$.

Hence $\frac{b^2}{\sqrt{a^2\cos^2\theta + a^2\sin^2\theta}} = c \Rightarrow b^2 = ac \Rightarrow$ a, b, c are in G.P.

79. A

Sol. The given lines intersects each other if $\begin{vmatrix} 2-1 & 3-4 & 4-5 \\ 1 & 1 & \lambda \\ \lambda & 2 & 1 \end{vmatrix} = 0$

$\Rightarrow \lambda = 0, -1$.

80. B

SECTION – B

81. 00008.00

Sol. Required number of arrangements

$= 4! - \left({}^2C_1 \cdot 3! \cdot 2! - {}^2C_2 \cdot 2! \cdot (2!)^2\right) = 8$.

82. 00005.00

Sol. Let Q(tan C, cot C), then

$\overrightarrow{PQ} = (\tan C - \sin A)\,\hat{i} + (\cot C - \cot B)\,\hat{j}$

$|\overrightarrow{PQ}|^2 = (\tan C - \sin A)^2 + (\cot C - \cot B)^2$

$|\overrightarrow{PQ}| = \overrightarrow{OQ} - \overrightarrow{OP}$

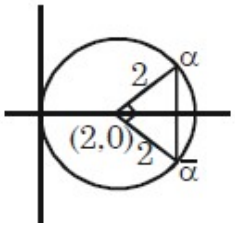
$|\overrightarrow{PQ}|^2 = \sqrt{\tan^2 C + \cot^2 C} - 1$

$= \left(\sqrt{(\tan^2 C - \cot^2 C)^2 + 2} - 1\right)^2$

min(PQ) = $\left(\sqrt{2} - 1\right)^2 = 3 - 2\sqrt{2}$.

83. 00006.82

Sol. $|\alpha-2|=2 \; and \; \overline{\alpha}-2=i(\alpha-2)$



$\Rightarrow \arg\dfrac{\overline{\alpha}-2}{\alpha-2}=\dfrac{\pi}{2}$

$area=\dfrac{1}{2}.2.2=2$

84. 00007.00

Sol. Let common difference be λ .

$\Rightarrow a(a+\lambda)=2a+5\lambda-1$

$=\lambda=\dfrac{16}{5-a}-a-3$

Possible values of 'a' are 3 and 4

85. 00002.00

86. 00002.00

Sol. $1920=5\times3\times2^7=5\times3\times2\times2\times2\times2\times2\times2\times2=\dfrac{9!}{7!}=9\times8$

$1920=5\times3\times4\times2\times2\times2\times2\times2\times1=\dfrac{9!}{5!}=9\times8\times7\times6$

$1920=3,5,4,4,2,2,2,1,1$

$\dfrac{9!}{2!2!3!}$

$3,5,4,4,4,2,1,1,1$

$\dfrac{9!}{3!3!}$

Total = $\dfrac{9!}{7!}+\dfrac{9!}{5!}+\dfrac{9!}{2!2!3!}+\dfrac{9!}{3!3!}$

$=\dfrac{9!}{7!}(1+42+210+140)$

$=393\ {}^{9}P_2$

87. 00005.00

Sol. $20c_0\times30c_{5m}+20c_1\times30c_{5m-1}$

$+.....+20c_{5m}\times30c_0$

Coeff of $x^{5m} \to (1+x)^{20}(1+x)^{30}$

$50c_{5m}, 5m = \frac{50}{2} = 25$

$m = 5$

88. 00012.00

Sol. $f(x)+f(x+2)=\sqrt{3}f(x+1)$

$x \to x+2 ------(1)$

$f(x+2)+f(x-14)=\sqrt{3}f(x+3) ------(2)$

$(1)+(2)$

$f(x)+2+(x+2)+f(x+14)$

$=\sqrt{3}\{f(x+1)+f(x+3)\}$

$=3f(x+2)$

$f(x)+f(x+4)=f(x+2)$

$x \to x+2 -----(3)$

$f(x+2)+f(x-16)=f(x+4) ---(4)$

$(3)+(4)$

$f(x)+f(x+6)=0$

$f(x+6)=-f(x)$

$f(x+12)=-f(x+6)$

$=f(x)$

Period $T=12$

89. 00019.00

90. 00081.00

Sol. Given $\frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}} \geq x$

$\frac{2f(x)f'(x)}{\sqrt{1-(f(x))^4}} - 2x \geq 0$

Let $g(x)=\sin^{-1}(f(x))^2 - x^2$ is non decreasing function

$\lim_{x\to a^-} g(x) \leq \lim_{x\to b^+} g(x)$

$\frac{\pi}{2} - a^2 \leq \frac{\pi}{6} - b^2$

$a^2 - b^2 \geq \frac{\pi}{3}; \left[a^2 - b^2\right] = 1$